

dply-example

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`subset()` and **dplyr** are both used to filter and manipulate data in R, but they differ in syntax, flexibility, and performance.

Sample Dataset

```
student <- data.frame(
  RollNo = c(101,102,103,104,105),
  Name = c("Rahul","Anita","Ajay","Sneha","Kiran"),
  Branch = c("CSE","ECE","CSE","IT","CSE"),
  Marks = c(85,90,76,95,88)
)

student
```

```
student <- data.frame(
  RollNo = c(101,102,103,104,105),
  Name = c("Rahul","Anita","Ajay","Sneha","Kiran"),
  Branch = c("CSE","ECE","CSE","IT","CSE"),
  Marks = c(85,90,76,95,88)
)

student
```

```
##   RollNo  Name Branch Marks
## 1    101 Rahul    CSE     85
## 2    102 Anita    ECE     90
## 3    103  Ajay    CSE     76
## 4    104 Sneha     IT     95
## 5    105 Kiran    CSE     88
```

1. Selecting Rows

Using `subset()`

```
subset(student, Branch=="CSE")
```

```
## RollNo Name Branch Marks
## 1 101 Rahul CSE 85
## 3 103 Ajay CSE 76
## 5 105 Kiran CSE 88
```

Using dplyr

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
## filter, lag
```

```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
student %>%
  filter(Branch=="CSE")
```

```
## RollNo Name Branch Marks
## 1 101 Rahul CSE 85
## 2 103 Ajay CSE 76
## 3 105 Kiran CSE 88
```

Output

Same result.

2. Selecting Columns

Using subset()

```
subset(student,
  select=c(Name,Marks))
```

```
## Name Marks
## 1 Rahul 85
## 2 Anita 90
## 3 Ajay 76
## 4 Sneha 95
## 5 Kiran 88
```

Using dplyr

```
student %>%  
  select(Name,Marks)
```

```
##      Name Marks  
## 1 Rahul      85  
## 2 Anita      90  
## 3  Ajay      76  
## 4 Sneha      95  
## 5 Kiran      88
```

3. Multiple Conditions

Using subset()

```
subset(student,  
        Branch=="CSE" &  
        Marks>80)
```

```
##   RollNo  Name Branch Marks  
## 1    101 Rahul    CSE     85  
## 5    105 Kiran    CSE     88
```

Using dplyr

```
student %>%  
  filter(Branch=="CSE",  
         Marks>80)
```

```
##   RollNo  Name Branch Marks  
## 1    101 Rahul    CSE     85  
## 2    105 Kiran    CSE     88
```

4. Sorting Data

Using Base R

```
student[order(student$Marks),]
```

```
##   RollNo  Name Branch Marks
## 3    103  Ajay   CSE    76
## 1    101 Rahul   CSE    85
## 5    105 Kiran   CSE    88
## 2    102 Anita   ECE    90
## 4    104 Sneha   IT     95
```

```
student[order(-student$Marks),]
```

Using dplyr

```
student %>%
  arrange(Marks)
```

```
##   RollNo  Name Branch Marks
## 1    103  Ajay   CSE    76
## 2    101 Rahul   CSE    85
## 3    105 Kiran   CSE    88
## 4    102 Anita   ECE    90
## 5    104 Sneha   IT     95
```

5. Adding a New Column

Base R

```
student$Result <- ifelse(student$Marks>=40,
                          "Pass",
                          "Fail")
```

Using dplyr

```
student %>%
  mutate(Result=
    ifelse(Marks>=40,
           "Pass",
           "Fail"))
```

```
##   RollNo  Name Branch Marks Result
## 1    101 Rahul   CSE    85   Pass
## 2    102 Anita   ECE    90   Pass
## 3    103  Ajay   CSE    76   Pass
## 4    104 Sneha   IT     95   Pass
## 5    105 Kiran   CSE    88   Pass
```

6. Grouping Data

Average marks branch-wise.

Base R

```
aggregate(Marks~Branch,  
          data=student,  
          mean)
```

```
##   Branch Marks  
## 1    CSE     83  
## 2    ECE     90  
## 3     IT     95
```

Using dplyr

```
student %>%  
  group_by(Branch) %>%  
  summarise(  
    Average=mean(Marks)  
  )
```

```
## # A tibble: 3 × 2  
##   Branch Average  
##   <chr>     <dbl>  
## 1 CSE         83  
## 2 ECE         90  
## 3 IT          95
```

7. Multiple Operations

Suppose we want

- Select CSE students
- Marks >80
- Display only Name and Marks
- Arrange by Marks

Base R / subset()

```
x <- subset(student,
             Branch=="CSE" &
             Marks>80)

x <- x[,c("Name", "Marks")]

x <- x[order(-x$Marks),]

x
```

```
##      Name Marks
## 5 Kiran     88
## 1 Rahul     85
```

Using dplyr

```
student %>%
  filter(Branch=="CSE",
         Marks>80) %>%
  select(Name, Marks) %>%
  arrange(desc(Marks))
```

```
##      Name Marks
## 1 Kiran     88
## 2 Rahul     85
```

Notice how the `dplyr` version reads almost like a sequence of English instructions.
